



RUNS

4 cached

+ New run

bench_4c19e0a2

benchmark

syn_7f3ac21e

fog 0.60

syn_2b90d411

fog 0.30

syn_e71c0a58

dark 0.70

new run

New run

Configure once; the same settings always produce the same experiment ID and cached results.

Real-Weather Benchmark

Score against SeeingThroughFog labels by condition and class.

Synthetic Degradation

Clean vs degraded on the sample video, your MP4, or clear dataset frames.

Dataset folder

~/data/SeeingThroughFog

Browse

Re-check

✓ cam_stereo_left_lut · 8-bit camera

✓ gt_labels · KITTI 2D boxes

✓ labeltool_labels · weather & light

✓ checksums · verified

Subset

seed 1337 · cap 300 / condition · 23 frames excluded

condition

frames

objects

smallest class

Clear · day

300

1,842

Ridable 112

Clear · night

300

1,215

Ridable 64

Fog · day

300

1,104

Ridable 58

Fog · night

212

688

Ridable 31 · low n

Snow

300

1,390

Ridable 71

Rain

300

1,262

Ridable 69

Fog chamber

180

402

Ridable 38 · low n

experiment bench_4c19e0a2 · cached, results reused

Reset

Run benchmark · 1,892 frames

REQUEST

POST /api/experiments

```
{
  "mode": "benchmark",
  "dataset": "~/data/SeeingThroughFog",
  "subset": { "seed": 1337, "cap": 300 },
  "model": "yolo-small.onnx",
  "provider": "cpu",
  "conf_floor": 0.10,
  "iou": 0.50,
  "class_map": "v1"
}
```

→ 202 Accepted · job id · progress over SSE

CLASS MAPPING V1

car	PassengerCars
truck, bus	LargeVehicles
bicycle, moto	RidableVehicles
person	Pedestrians
Vehicle, Obstacle	ignored

Q fog

esc

Run fog 0.60 on Clear · day frames

cached

Sweep fog severity 0.2 → 1.0

5 runs

Go to Fog · day benchmark

g b

Open sim-to-real finding

g f

Perception Reliability Lab

experiments / bench_4c19e0a2 / fog · day / #0142

benchmark · 55%

Search frames, runs, commands

127.0.0.1:8000

Export report

EXPLORER

1,892 frames

Clear · day

.61

Clear · night

.49

Fog · day

.42

#0233

3.25

#0519

2.75

#0142

2.25

#0088

2.00

#0376

1.75

+ 295 more

Fog · night

.34

Snow

.45

Rain

.52

Fog chamber

.43

Synthetic fog 0.60

.52

#0142 · fog · day

#0233 · fog · day

sim-to-real

Labels

Predictions

Ignore regions

threshold 0.40

FALSE ALARM car .43

MISS RidableVehicles

HIT person .52 car .71

CLASS × car .48

← → frame · L labels · P predictions · ⌘ commands

2 unmapped detections not scored

Job

Frame table

job 7c1e · 1,040 / 1,892

Cancel

18:42:07 job 7c1e started · 1,892 frames · CPUExecutionProvider · 3 warm-ups excluded

18:42:31 clear · day 300/300 p50 38.1 ms done

18:42:54 clear · night 300/300 p50 38.3 ms done

18:43:02 fog · day 140/300 p50 38.2 ms streaming...

18:43:02 warn 3 frames: unmapped classes only (traffic light, stop sign)

18:43:03 event: progress {"done":1040,"total":1892,"condition":"fog_day"}

INSPECTOR

SELECTED · CLASS CONFUSION

LargeVehicles → car 0.48

IoU

0.83

box (norm.)

.69 .43 .21 .27

label src

gt_labels

score weight

0.75

FRAME #0142

Hits

2

Misses

1

False alarms

1

Error score

2.25

AP · CLEAR → FOG · DAY

Passenger

.78 → .58

Large

.61 → .44

Ridable

.42 → .24

Pedestrians

.64 → .41

Open findings →

v0.3

stf_subset_s1337.json

YOLO small · ONNX · CPU

Perception Reliability Lab

experiments / syn_7f3ac21e / frame 17

cached

Search frames, runs, commands

⌘K

127.0.0.1:8000

Export report

SYNTHETIC RUNS

sample_intersection.mp4

fog 0.6070%

fog 0.3088%

darkness 0.7064%

jpeg 0.8079%

clear · day frames

fog 0.60mAP .52

+ new condition

sample_intersection.mp4 · fog 0.60

clear · day · fog 0.60

Detections

Sync frames

threshold0.250.400.60

CLEAN BASELINE · 5 shown

DEGRADED · FOG 0.60 · 3 shown · 1 dropped

traffic light .68

bicycle .57

car .91

truck .83

stop sign .88

frame 17 / 24 · t = 8.00 s

instability per frame · darker = worse

dropped traffic light

car .74

NEW car .44 .52 · was truck

Frame 17 matches

Job

clean	degraded	IoU	Δ conf	outcome
car .91	car .74	0.88	-0.17	retained
truck .83	bus .52	0.81	-0.31	class change
stop sign .88	stop sign .61	0.79	-0.27	retained
bicycle .57	bicycle .31	0.72	-0.26	retained · below threshold
traffic light .68	-	-	-	dropped
-	car .44	-	-	introduced

% = retention vs clean baseline

unlabeled video: stability only

DEGRADATION

ONE PER EXPERIMENT

Darkness

Blur

Fog

Noise

JPEG

severity0.60

transmission0.58

airlight0.85

seed1337

STABILITY VS CLEAN

Retention70%102 / 146

Introduced12%14 / 119

Class changes3truck → bus

Conf. shift-0.210.77 → 0.56

Consistency, not accuracy: this clip has no labels.

LATENCY

inference p5038.9 ms

effective fps25.7

v0.3

01 · Sim-to-real

02 · Where it fails

03 · Precision–recall

04 · Worst frames

CONDITIONS

mAP

Clear · day	.61
Clear · night	.49
Fog · day	.42
Fog · night	.34
Snow	.45
Rain	.52
Fog chamber	.43
Synthetic fog 0.60	.52

findings · bench_4c19e0a2

FINDING 01 · SIM-TO-REAL

Synthetic fog explains about half of how real fog breaks the detector.

Clear · day

0.61

mAP · 1,842 objects

Synthetic fog 0.60

0.52

same 300 frames

Real fog · day

0.42

mAP · 1,104 objects

■ AP drop · synthetic

■ AP drop · real

PassengerCars		55% reproduced
LargeVehicles		53% reproduced
RidableVehicles		39% reproduced
Pedestrians		39% reproduced

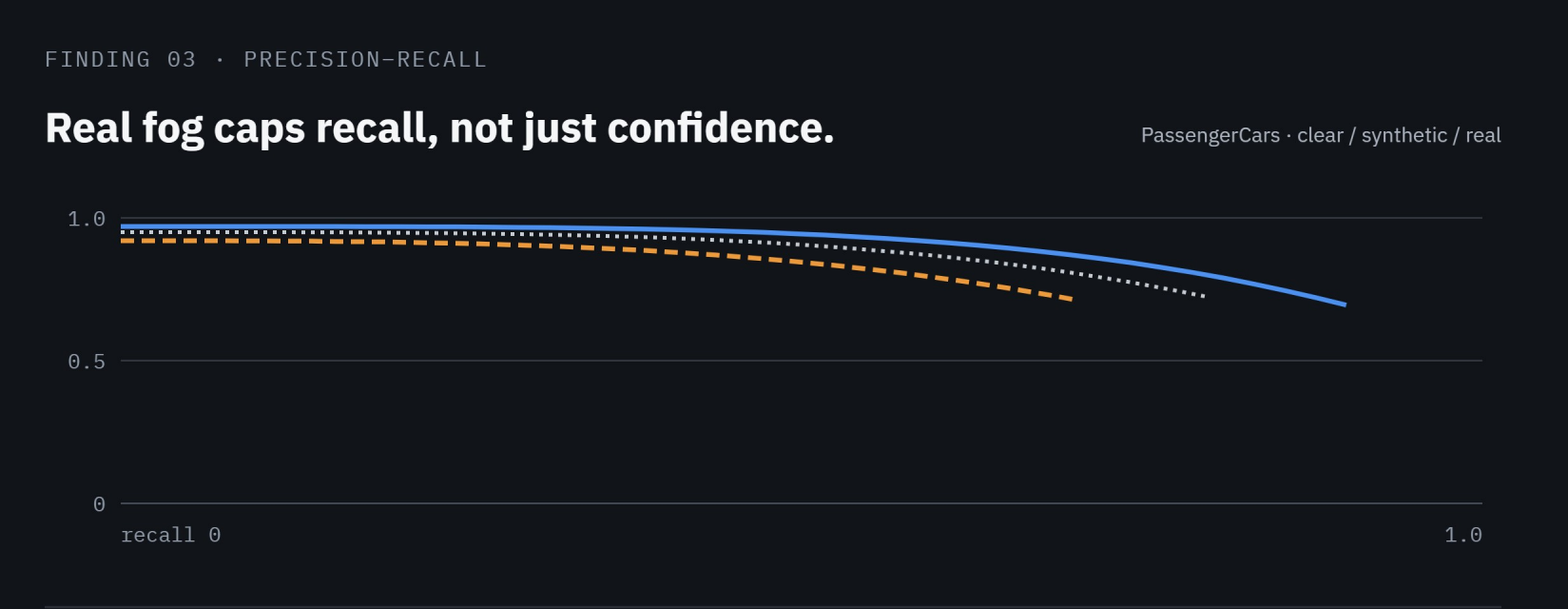
Distributional, not paired: real-fog frames are different scenes. Both sides are daytime to limit the lighting confound.

FINDING 02 · WHERE IT FAILS

Ridable vehicles are the weak spot in every condition.

	Passenger	Large	Ridable	Pedestr.	mAP
Clear · day	0.78	0.61	0.42	0.64	0.61
Clear · night	0.66	0.52	0.30	0.47	0.49
Fog · day	0.58	0.44	0.24	0.41	0.42
Fog · night	0.49	0.36	0.19 *	0.33	0.34
Snow	0.61	0.47	0.26	0.45	0.45
Rain	0.69	0.55	0.33	0.52	0.52
Fog chamber	0.55	—	0.28 *	0.46	0.43
Synthetic fog 0.60	0.67	0.52	0.35	0.55	0.52

AP @ IoU 0.50 · * fewer than 50 objects



04 · WORST FRAMES

#1 · #0233

Fog · day3.25

#2 · #0877

Fog · night3.00

#3 · #0413

Snow2.50

#4 · #1206

Fog chamber2.00

score = 1.00·misses + 0.50·false alarms + 0.75·confusions · images shown locally, excluded from exports

RUN

BENCH_4C19E0A2

complete · 2 m 14 s

frames	1,892
objects	7,903
manifest	stf_subset_s1337
model	YOLO small · ONNX
class map	v1 · ignore rule
match	IoU ≥ 0.50

LATENCY

read p50	4.6 ms
inference p50	38.2 ms
inference p90	44.0 ms
effective fps	26.2

session.run only · 3 warm-ups excluded

- READ THIS FIRST
- Accuracy on this dataset, subset and class mapping only.
 - Weather co-occurs with lighting, road and traffic changes.
 - Not validated for safety-critical or operational use.
- [Export as report →](#)

EXPORTS

draft · bench_4c19e0a2

now

syn_7f3ac21e.html

sep 26

syn_2b90d411.html

sep 25

v0.3

exports/ · local only

